Kinyarwanda News Topic Classification: Comparing Classical, Recurrent, and African-Centric Transformer Models on KINNEWS

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Machine Learning Techniques 2

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GitHub Repository: Link

Video: Link

Web App: Link

Hugging Face App: Link

Abstract

I built and deployed a system that classifies Kinyarwanda news articles into 14 topics, and compared three families of models on the KINNEWS corpus (Niyongabo et al., 2020). Auditing the data showed that over half of the 17,014 training rows were exact duplicates, that 156 articles carried conflicting labels, and that the class distributions of the official train and test splits differ sharply. After deduplication, I trained classical baselines (TF-IDF with logistic regression), BiLSTM models with Word2Vec embeddings and attention, and two African-centric pretrained Transformers, AfriBERTa (Ogueji et al., 2021) and AfroXLMR (Alabi et al., 2022). Class weighting was the most effective single change, raising the baseline's macro-F1 from 0.458 to 0.577. Fine-tuned Transformers only modestly beat this well-tuned baseline: the model chosen on validation (AfriBERTa, 512 tokens) reached 75.5% accuracy and 0.581 macro-F1 on the deduplicated test set, while AfroXLMR reached 77.3% and 0.618. Error analysis traced most remaining errors to rare and overlapping categories, a source-dependent split, and likely label noise, rather than to shortcut features. The final model is served through a public Streamlit web app.

1. Introduction

The objective of this project is to build an accurate, deployable topic classifier for Kinyarwanda news, and to explain which modelling choices matter for a low-resource African language. Kinyarwanda is the national language of Rwanda and is spoken by most of its population, yet it remains low-resourced in NLP, with few labelled datasets and models (Niyongabo et al., 2020).

Rwandan news sites publish a large volume of Kinyarwanda content every day. Automatic topic classification is a basic building block for organising it: news aggregation, search, recommendation and media monitoring all depend on it. Most off-the-shelf classification tools, however, are trained for high-resource languages and transfer poorly. The target users of this system are Rwandan news platforms and aggregators that need to sort incoming articles, media-monitoring and research organisations that track coverage by topic, and NLP researchers who need a Kinyarwanda baseline.

Kinyarwanda is also technically interesting. It is an agglutinative Bantu language: one word can carry subject, tense, object, and verb root, as in *ntibazabibona* ("they will not see them"). This creates many rare word forms, which makes word-level statistics sparse (Nzeyimana and Niyongabo Rubungo, 2022). I chose the language because I speak it, which lets me judge model errors and data quality directly, and I chose topic classification because a public labelled dataset exists, so the effort could go into modelling and analysis.

The project addresses four questions:

1. How clean and representative is the main public dataset, KINNEWS?
2. Do neural sequence models beat strong classical baselines on this task?
3. Do Transformers pretrained on African languages help, and which pretraining works best for Kinyarwanda?
4. What kinds of errors remain, and why?

2. Related Work

Kinyarwanda text classification rests mainly on one benchmark, KINNEWS, and recent progress has come from Transformers pretrained on African languages; data quality and tokenizer fit have received less attention.

Kinyarwanda news classification. Niyongabo et al. (2020) introduced KINNEWS and its Kirundi counterpart KIRNEWS, collected from news websites in each country. They benchmarked classical classifiers and neural models built on pretrained word embeddings, and studied cross-lingual transfer from Kinyarwanda to Kirundi. Their work established the dataset I use and the classical-versus-neural comparison I revisit.

Pretrained models for African languages. Large multilingual models such as XLM-R (Conneau et al., 2020) cover 100 languages but include few African ones. AfriBERTa (Ogueji et al., 2021) took the opposite route: a smaller model pretrained from scratch on under 1 GB of text in 11 African languages, including Kinyarwanda, which matched or beat larger multilingual models on classification and named-entity recognition. AfroXLMR (Alabi et al., 2022) instead adapted XLM-R through multilingual adaptive fine-tuning, continuing its pretraining on 17 African languages. KinyaBERT (Nzeyimana and Niyongabo Rubungo, 2022) added explicit morphological analysis for Kinyarwanda and reported gains on several downstream tasks, which underlines how much Kinyarwanda morphology matters.

African news topic classification more broadly. MasakhaNEWS (Adelani et al., 2023) benchmarked news topic classification for 16 African languages. Fine-tuned African-centric pretrained models generally performed best there, but simple baselines remained competitive for several languages.

Shortcut learning. Models often exploit spurious cues that correlate with labels in the training data, such as a source name, rather than the intended signal (Geirhos et al., 2020). News corpora assembled from a few websites are especially exposed to this.

Is the problem already solved? I found no public, deployed Kinyarwanda news topic classifier at the time of writing. KINNEWS released benchmark results but no usable system, and MasakhaNEWS covers Kirundi but not Kinyarwanda (Adelani et al., 2023).

The gap this project addresses. Previous work reports scores on KINNEWS as distributed. I instead audit the dataset first (duplicates, conflicting labels, split shift), evaluate on a deduplicated test set, compare classical, recurrent, and two kinds of African-centric Transformer under the same protocol, test the shortcut hypothesis directly, measure how well each tokenizer fits Kinyarwanda, and deploy the result as a usable application.

3. Dataset and Data Preparation

After auditing KINNEWS, I kept 6,726 training, 745 validation, and 3,464 test articles in 14 categories. The audit removed more than half of the raw training rows as duplicates and revealed a strong train/test distribution shift.

3.1 Source

KINNEWS (Niyongabo et al., 2020) contains Kinyarwanda news articles collected from Rwandan news websites, each with a URL, title, body, and one of 14 topic labels. It is distributed under the MIT license with official train (17,014 rows) and test (4,254 rows) splits, in raw and pre-cleaned versions. The 14 categories (politics, sport, economy, health, entertainment, history, technology, tourism, culture, fashion, religion, environment, education, relationship) were defined by the dataset's creators, and I used them unchanged. I used the raw version because the cleaned version removes stopwords and casing that pretrained Transformers can use. No synthetic or machine-generated data was used anywhere in this project. Since the Hugging Face copy depends on a loading script that current library versions no longer run, notebook 01 downloads the original archive directly.

The dataset suits the task: it is the largest public labelled Kinyarwanda news corpus, its articles are real published news, and its categories match how Rwandan outlets organise content.

3.2 Data quality audit

Check	Finding	Action
Exact duplicate articles (train)	17,014 rows contain only 7,630 unique texts; 9,387 repeated URLs; one article appears 80 times	Kept one copy of each
Conflicting labels	156 texts appear with more than one label	Removed (the correct label is unknowable)
Train/test overlap	3 test articles also in training	Removed from training; test kept intact
Duplicates (test)	4,254 rows contain 3,464 unique texts	Evaluated on the deduplicated test set; original also logged
Very short texts	1 test article under 5 words	Kept (reported only)

Duplicates matter because a model trained on the raw data would see some articles up to 80 times, which skews what it learns, and duplicated test articles would count some predictions several times. Results on the original 4,254-row test set are logged in the repository for comparison with earlier work.

3.3 Final splits and the distribution shift

I held out a stratified 10% of the cleaned training pool for validation (random seed 42), used for early stopping and model selection. The test set was used once per experiment.

Category	Train	Val	Test	Train %	Test %
politic	1,591	176	315	23.7	9.1
sport	878	98	744	13.1	21.5
economy	998	111	174	14.8	5.0
health	976	108	272	14.5	7.9
entertainment	900	98	774	13.4	22.3
religion	401	44	256	6.0	7.4
relationship	380	42	340	5.6	9.8
history	196	22	11	2.9	0.3
technology	135	15	103	2.0	3.0
education	104	12	70	1.5	2.0
culture	69	8	138	1.0	4.0
fashion	34	4	91	0.5	2.6
environment	34	4	100	0.5	2.9
tourism	30	3	76	0.4	2.2
Total	**6,726**	**745**	**3,464**	100	100

Two properties of these splits shape every result in this report. First, the imbalance is severe: *politics* has 53 times more training articles than *tourism*. Second, the train and test distributions differ sharply: *environment*, *fashion* and *tourism* are five to six times more common in test than in training, while *history* is ten times rarer. Because the validation set is drawn from the training pool, it follows the training distribution and contains only 3 or 4 articles for some classes. The shift also appears to follow news sources: the site name *muhabura* appears in about 40% of training articles in economy, health and history, but in almost none of the test articles in those

classes (Section 6.4).

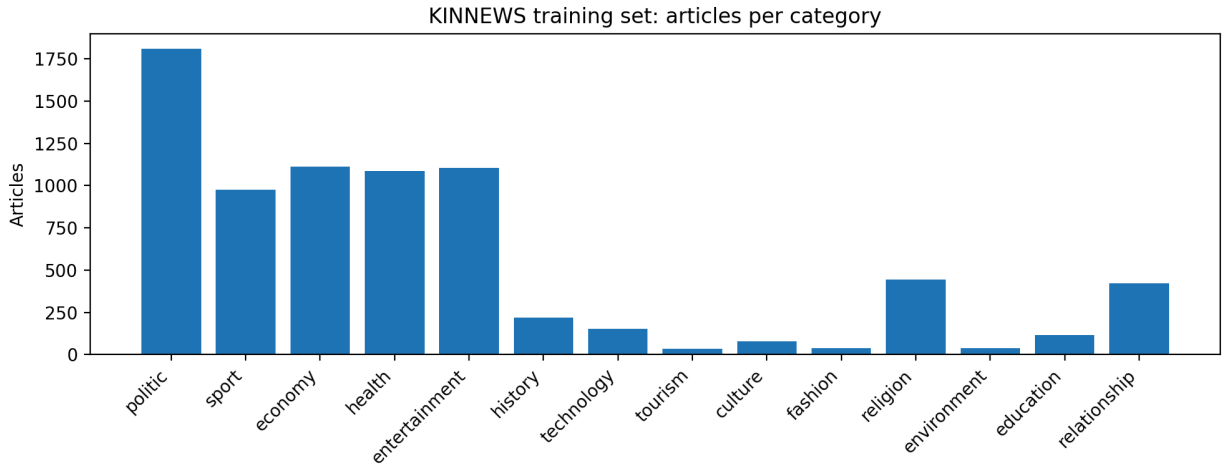


Figure 1: training articles per category

3.4 Text characteristics and preprocessing

Articles are long: the median is 292 words (mean 346), 10% exceed 573 words, and the longest has 11,583. The 6,726 training articles contain 140,349 distinct word forms, a sign of rich morphology and spelling variation. Even with a 50,000-word vocabulary, 7.8% of test tokens were never seen in training.

Preprocessing was deliberately light, because pretrained tokenizers expect natural text:

- Decoding HTML entities and normalising Unicode (NFC);
- **Standardising apostrophes** (curly to straight), because Kinyarwanda contractions such as *n'*, *y'* and *by'* are very frequent and inconsistent characters would split one word into several tokens;
- Removing URLs and e-mail addresses and collapsing whitespace;
- Joining title and body, since the headline is often the most informative part.

Model-specific steps (lower-casing, tokenisation, truncation) are described with each model in Section 4.

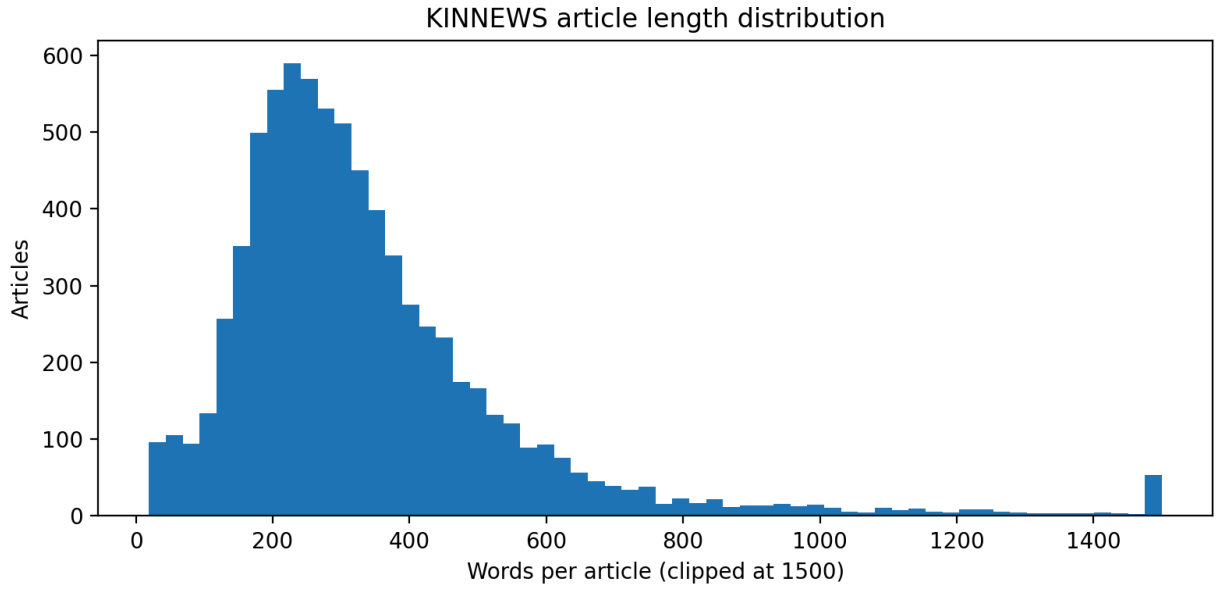


Figure 2: article length distribution

4. Methodology

I compared three model families of increasing complexity under one protocol: hyperparameters and the final model were chosen on validation macro-F1, and each experiment changed exactly one thing from the one before it.

4.1 Evaluation protocol

The primary metric is **macro-F1**, the unweighted mean of the per-class F1 scores. With this level of imbalance, accuracy is dominated by the largest classes and would hide failures on rare ones. I also report accuracy, weighted-F1, and per-class precision and recall. All experiments use random seed 42 and are run on a Google Colab T4 GPU.

4.2 Classical baselines (B0–B3)

- **B0, majority class:** always predicts the most frequent class, as a lower bound.
- **B1, word TF-IDF + logistic regression:** unigrams and bigrams, lower-cased, minimum document frequency 2, maximum 90%, log-scaled term frequency, up to 200,000 features.
- **B2, character TF-IDF + logistic regression:** character 2- to 5-grams within word boundaries, up to 300,000 features. The motivation is morphology: character n-grams let different inflections of the same root share features.

- **B3, B2 + balanced class weights:** each class's loss is weighted inversely to its frequency.

4.3 BiLSTM models (N1–N3)

Text is lower-cased and split into alphabetic tokens, which separates contractions (*n'umwana* → *n*, *umwana*), so a noun gets the same token with or without an attached particle. The vocabulary holds the 50,000 most frequent training words seen at least twice, plus padding and unknown tokens. Sequences are truncated to 400 tokens, which covers most articles in full.

The network is: a 200-dimensional embedding layer, then a bidirectional LSTM (Hochreiter and Schmidhuber, 1997) with 128 units per direction, then pooling over time, dropout of 0.3, and a linear layer producing 14 class scores. Packed sequences let the LSTM skip padding, and padding is masked out of the pooling.

- **N1:** randomly initialised embeddings, max pooling (the strongest signal per dimension anywhere in the article).
- **N2:** embeddings initialised from Word2Vec (Mikolov et al., 2013), trained with gensim (Řehůřek and Sojka, 2010) on the training split only: skip-gram, 200 dimensions, window 5, minimum count 2, 10 epochs, giving 63,430 word vectors.
- **N3:** N2 with additive attention pooling (Yang et al., 2016). A learned layer scores each position; the scores are normalised with softmax, and the article is represented by the weighted sum of LSTM states. The weights also show which words the model focused on.

Training used AdamW (learning rate 1e-3, weight decay 1e-4), batch size 32, gradient clipping at 1.0, and early stopping after 3 epochs without validation improvement (at most 15 epochs). These models used an unweighted loss, so their fair comparison is B2 rather than B3.

4.4 Fine-tuned Transformers (T1–T3)

I fine-tuned two pretrained models that share the XLM-R architecture but differ in pretraining:

Model	Parameters	Pretraining	Vocabulary
AfriBERTa-base (`castorini/afriberta_base`)	~111M	From scratch on 11 African languages incl. Kinyarwanda	70,006 sub-words
AfroXLMR-base (`Davlan/afro-xlmr-base`)	~270M	XLM-R adapted to 17 African languages incl. Kinyarwanda	250,002 sub-words

A new classification head (dense layer + output layer on the first-token representation) replaces each model's pretraining head. Fine-tuning used AdamW (learning rate 3e-5, weight decay

0.01) with linear warm-up over the first 10% of steps, then linear decay. Warm-up avoids large early updates that would damage the pretrained weights. Other settings were batch size 16, FP16 mixed precision, gradient clipping at 1.0, at most 6 epochs, and early stopping after 2 epochs without improvement. The loss was **class-weighted cross-entropy** with the same weights as B3, because class weighting had been the largest gain among the baselines. Articles were truncated from the end, which always keeps the title, and padded per batch.

- **T1:** AfriBERTa, 256 tokens.
- **T2:** AfroXLMR, 256 tokens (one change: the pretrained model).
- **T3:** the better of T1 and T2 on validation (AfriBERTa), 512 tokens, with a batch of 8 and 2 accumulation steps to keep the effective batch at 16 (one change: context length).

4.5 What I implemented and what I reused

Implemented: data download, cleaning, deduplication and the label-conflict audit; the splits; all training and evaluation code, including the BiLSTM with masked max and attention pooling and the Transformer fine-tuning loop with weighted loss, warm-up scheduling and mixed precision; the experiment design; the tokenizer, shortcut, confidence and length analyses; the manual error review; and the web application.

Reused: the KINNEWS dataset; the pretrained AfriBERTa and AfroXLMR checkpoints; and the scikit-learn (Pedregosa et al., 2011), gensim, PyTorch, Hugging Face Transformers (Wolf et al., 2020) and Streamlit libraries.

5. Experiments and Results

The best test results came from fine-tuned Transformers, but a class-weighted character TF-IDF model came within 0.004 macro-F1 of the validation-selected Transformer. The largest single gain came from class weighting, not from model complexity.

5.1 All experiments

Exp	Model	Change from previous	Val macro-F1	Test acc.	Test macro-F1	Test weighted-F1	Train time (min)
B0	Majority class	n/a	0.027	0.091	0.012	0.015	0
B1	Word TF-IDF + LogReg	n/a	0.553	0.691	0.413	0.656	3.7
B2	Char TF-IDF + LogReg	words → characters	0.623	0.710	0.458	0.681	7.5

Exp	Model	Change from previous	Val macro-F1	Test acc.	Test macro-F1	Test weighted-F1	Train time (min)
B3	B2 + balanced weights	class weighting	0.745	0.743	0.577	0.742	5.2
N1	BiLSTM, random emb., max pool	n/a	0.530	0.576	0.354	0.572	2.1
N2	BiLSTM, Word2Vec, max pool	embedding init.	0.544	0.595	0.383	0.594	1.4
N3	BiLSTM, Word2Vec, attention	pooling	0.631	0.685	0.513	0.687	1.7
T1	AfriBERTa, 256 tokens	n/a	0.686	0.748	0.578	0.742	7.0
T2	AfroXLMR, 256 tokens	pretrained model	0.665	**0.773**	**0.618**	**0.771**	15.0
T3	AfriBERTa, 512 tokens (final)	context length	**0.691**	0.755	0.581	0.751	13.5

Test results are on the deduplicated test set (3,464 articles). Train times exclude Word2Vec training (5.9 min) and model downloads. Macro precision and recall for the two final candidates were 0.659 and 0.611 for B3, and 0.631 and 0.622 for T3; per-class precision, recall, and F1 for every model are in the repository's results folder.

5.2 What each experiment showed

Character features beat word features (B1 → B2, +0.045). With 140,349 word forms in 6,726 articles, most words are rare. Character n-grams let *abanyeshuri*, *banyeshuri* and *abanyeshuli* share most of their features, which suits Kinyarwanda's morphology.

Class weighting was the largest gain (B2 → B3, +0.119). It mainly lifted rare classes, which are 5–6 times more frequent in the test set than in training. The validation-selected regularisation also changed (C = 10 → 1): once rare classes carry large weights, stronger regularisation likely prevents the model from overfitting their few examples.

Pretrained word embeddings helped the BiLSTM learn faster but only slightly better (N1 → N2, +0.029). Validation macro-F1 after one epoch rose from 0.16 to 0.33, and N1 was still improving at its 15-epoch limit. The nearest neighbours of the trained Word2Vec vectors were linguistically sensible and mostly captured spelling variants: *abanyeshuri* (students) sits next to *abanyeshuli* and *abarimu* (teachers); *amafaranga* (money) next to *amafranga*, *amafaraga* and *amafanga*; *perezida* next to *prezida* and *kagame*.

Attention pooling was the largest gain among the BiLSTMs (N2 → N3, +0.130). Max pooling picks up the strongest signal anywhere, including noise in long articles. Attention learns a weighted average focused on informative words such as *umuhanzi* (artist) or *fashion*. N3's training loss fell to 0.03 while validation plateaued, a clear sign of overfitting on this small dataset. Without class weights, N3 still beat its fair comparison B2 (0.513 vs 0.458), but not the weighted B3.

AfriBERTa's tokenizer fits Kinyarwanda better (T1 vs T2).

Tokenizer	Vocabulary	Sub-words per word	Median tokens per article	Truncated at 256	Truncated at 512
AfriBERTa-base	70,006	1.88	525	95%	54%
AfroXLMR-base	250,002	2.49	713	96%	77%

Despite a vocabulary 3.5 times smaller, AfriBERTa splits *Abanyeshuri* into *Abanyeshur* + *i*, whereas AfroXLMR splits it into *Aba* + *ny* + *es* + *huri*. AfriBERTa's vocabulary was learned from African-language text, while AfroXLMR inherits XLM-R's vocabulary, built for 100 mostly non-African languages. Yet AfroXLMR scored higher on the test set (0.618 vs 0.578). Its larger capacity and broader pretraining appear to outweigh its poorer tokenisation, at twice the training time.

Reading twice as much text barely helped (T1 → T3, +0.003). Although 95% of articles are cut at 256 tokens, the topic of a news article is usually clear from its title and opening paragraphs, which are always kept. Section 6.3 shows that longer articles are harder for every model, including the baseline that reads the full text, so the difficulty lies in the articles rather than in truncation.

Transformers beat a well-tuned baseline only modestly. T3 essentially ties B3 (0.581 vs 0.577), and the best Transformer (T2) is ahead by 0.041 macro-F1 and 3.0 accuracy points.

Topic classification is largely keyword-driven, which TF-IDF captures well, and the train/test shift limits all models regardless of architecture. This matches MasakhaNEWS, where simple baselines remained competitive for several African languages (Adelani et al., 2023).

5.3 Model selection: validation and test disagree

On validation, T3 was best (0.691); on test, T2 was best (0.618 vs 0.581). I kept **T3 as the final, deployed model** because it was chosen by the pre-defined validation protocol; switching to T2 after seeing test scores would use the test set for model selection and overstate the expected performance. The disagreement has two causes. The validation set contains only 3 or 4 articles for *tourism*, *fashion,* and *environment*, so a single changed prediction moves its macro-F1 noticeably. And validation follows the training distribution, not the shifted test distribution. A larger, stratified validation set is the obvious fix (Section 8). T3 is also less than half the size of T2, which makes it cheaper to serve on free hosting.

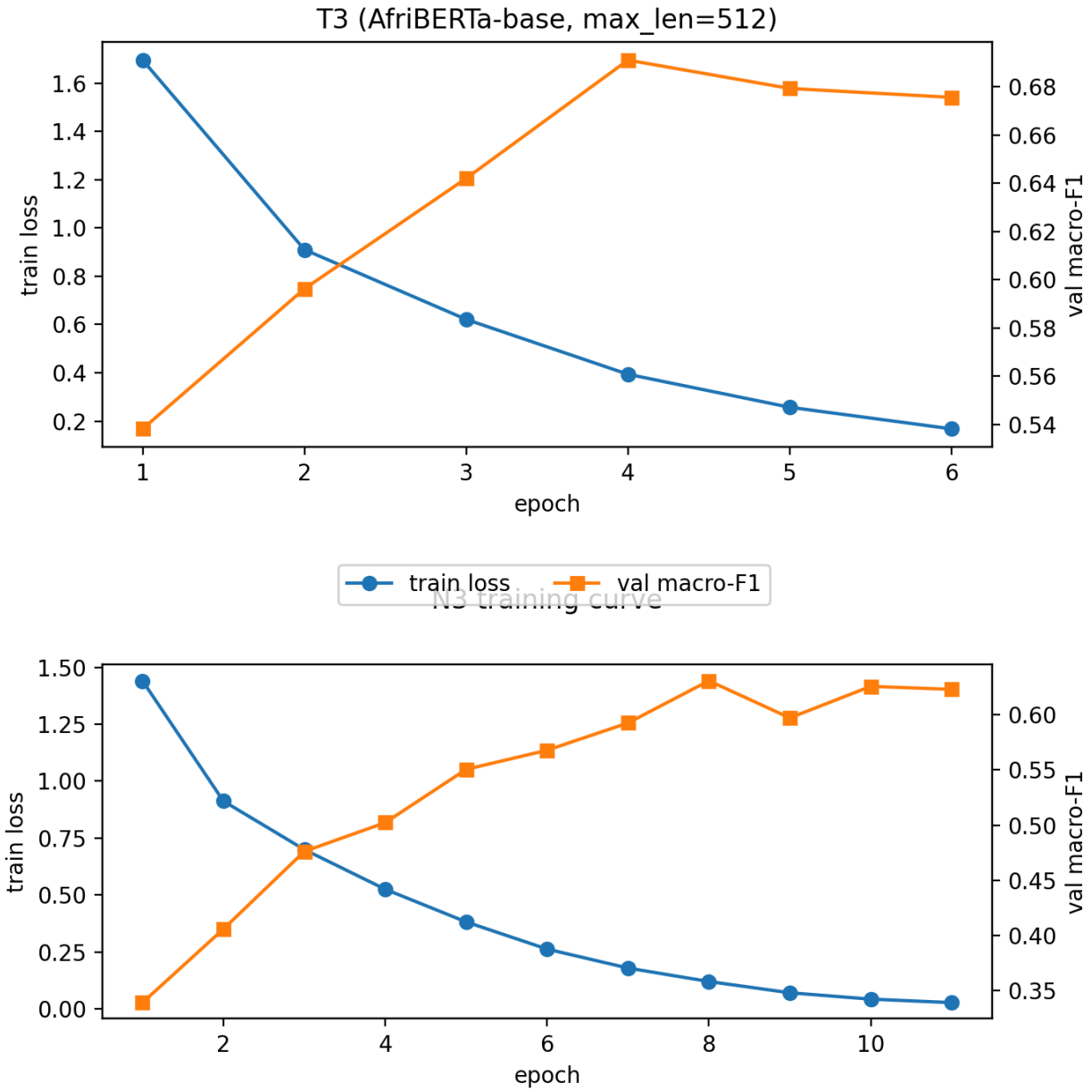


Figure 3: training curves figures

6. Error Analysis and Discussion

Most remaining errors come from the data rather than from model capacity: rare and overlapping categories, a source-dependent split, and likely mislabelled articles. The shortcut hypothesis I set out to test was not supported. All analyses compare the best model of each family (B3, N3 and T3) on the same 3,464 test articles.

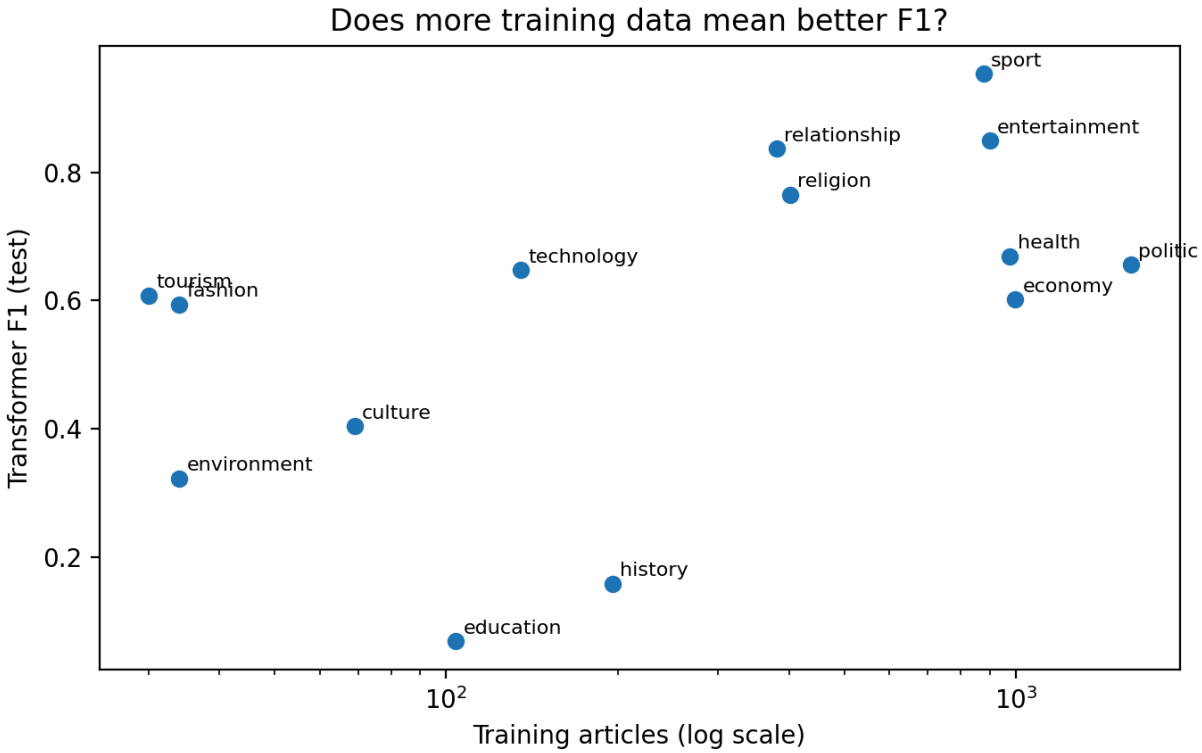
6.1 Which categories fail, and is it just data size?

Category	Train articles	Baseline B3 F1	BiLSTM N3 F1	Transformer T3 F1
politic	1,591	0.625	0.638	0.656
economy	998	0.558	0.482	0.602
health	976	0.656	0.492	0.669
entertainment	900	0.834	0.828	0.849
sport	878	0.946	0.919	0.953
religion	401	0.771	0.728	0.765
relationship	380	0.819	0.663	0.837
history	196	0.126	0.119	0.159
technology	135	0.674	0.607	0.648
education	104	0.109	0.240	0.069
culture	69	0.434	0.326	0.404
fashion	34	0.639	0.593	0.594
environment	34	0.378	0.142	0.323
tourism	30	0.513	0.408	0.608

Training size explains part of the pattern: the correlation between log training count and Transformer F1 is 0.52. The exceptions are more informative. *Tourism* and *fashion* score around 0.6 with only 30 - 34 training articles, because their vocabulary is distinctive: the baseline's top features include *pariki* (park), *rdb*, *nyungwe* and *ubukerarugendo* for tourism, and *miss*, *imideli* and *fashion* for fashion. *Education* and *history* fail with 104 - 196 articles, because their content overlaps with other categories (Section 6.2). *History* has only 11 test articles, so its F1 is also very unstable: the Transformer finds 9 of them but labels about 90 other articles as history,

giving a precision of 0.09.



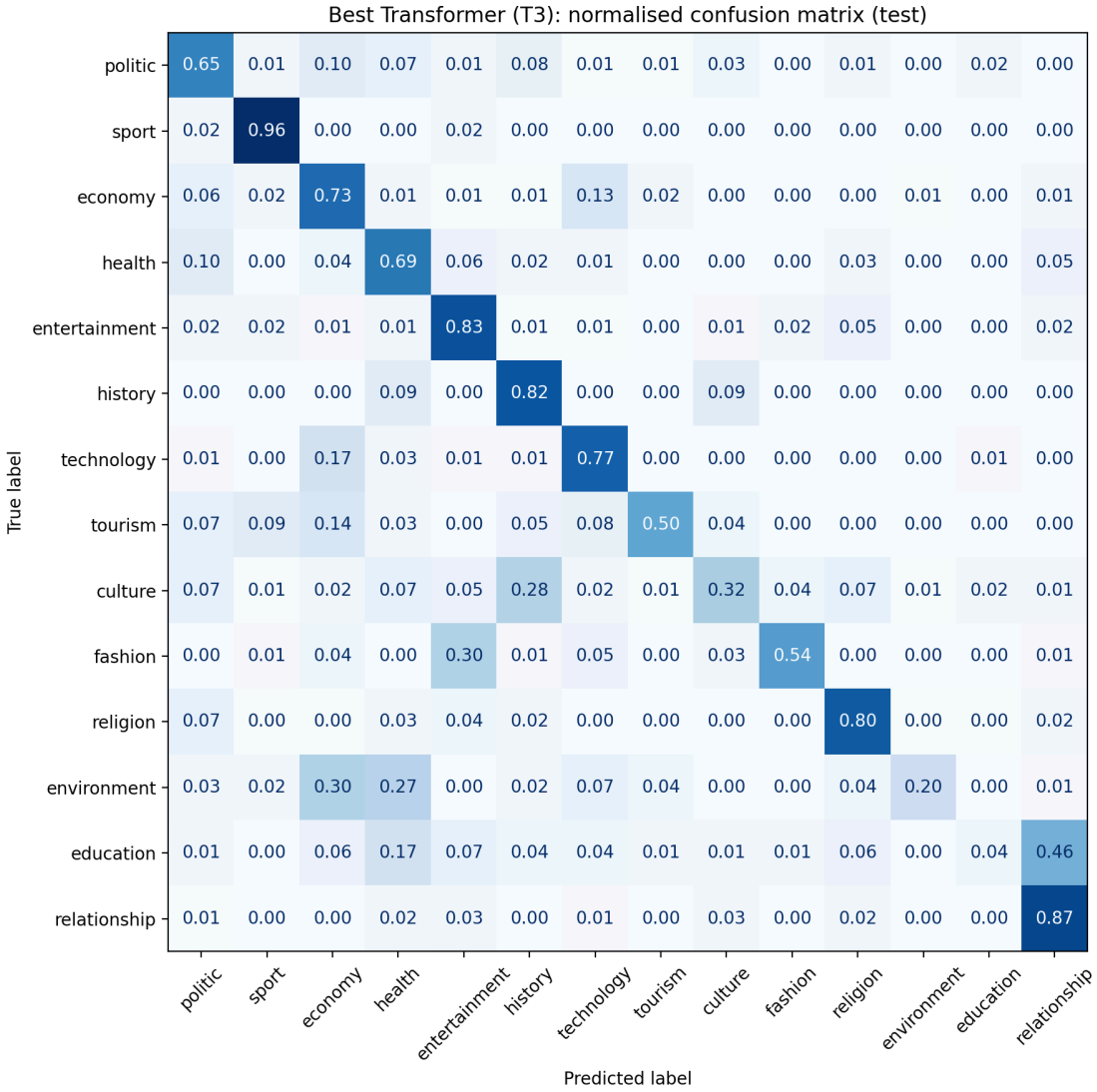


6.2 The most frequent confusions

The Transformer makes 850 errors (24.5%). The most frequent confusions are between categories whose content genuinely overlaps:

True → predicted	Errors	% of true class	Likely reason
education → relationship	32	45.7	articles on youth, students' lives or parenting
environment → economy	30	30.0	Agriculture, land and resource articles framed economically
fashion → entertainment	27	29.7	Miss Rwanda and celebrity coverage spans both
culture → history	39	28.3	Shared vocabulary of tradition and the past (*umwami*, *amateka*)
environment → health	27	27.0	Sanitation, water and pollution articles
economy → technology	23	13.2	Business articles about telecoms, banking and digital services
politic → economy	31	9.8	Government policy on budgets and development
politic → history	25	7.9	Genocide commemoration (*Kwibuka*, *jenoside*) covered under both labels

These are not random mistakes: in most cases an article could reasonably carry either label. Single-label topic classification forces one choice where the content is multi-topic.



6.3 Agreement, confidence and article length

Different models fail on the same articles. All three models are correct on 58.5% of test articles, and all three are wrong on 15.7% (545 articles). Only 3.4% are solved by the Transformer alone. Models with very different inductive biases (bag of character n-grams, recurrent network, pretrained Transformer) failing on the same articles points to the articles or their labels, not to a lack of model capacity.

Confidence is informative but not fully reliable.

Transformer confidence	Articles	Accuracy
above 90%	2,516	86.8%
70–90%	488	51.0%
50–70%	321	43.9%
below 50%	139	29.5%

Accuracy rises steadily with confidence, which justifies the low confidence warning in the web app. However, 333 errors were made with over 90% confidence, so a confident prediction is not proof of a correct one. Section 6.5 shows that some of these are probably label errors.

Long articles are harder for every model. Transformer accuracy falls from 79.4% for articles under 200 words to 59.2% for those over 600 words. The baseline falls almost identically (77.3% to 57.5%), although it reads the whole text. Truncation is therefore not the main cause; long articles tend to be features, interviews, or opinion pieces that cover several topics.

6.4 Testing the shortcut hypothesis

The baseline's highest weighted features included source and byline words such as *muhabura* (a news website), *yanditswe na* ("written by"), *src*, *caption,* and *views*. This suggested shortcut learning, where a model uses cues that correlate with labels instead of the topic itself (Geirhos et al., 2020). I tested it directly by deleting these words from every test article and predicting again without retraining:

Model	Original macro-F1	Markers removed	Change
Baseline B3	0.577	0.580	+0.003
Transformer T3	0.581	0.581	0.000

The hypothesis was not supported: removing the markers left both models unchanged, so their predictions rest on the article content. The test did reveal something else. The same markers are distributed in opposite ways across the splits. *Muhabura* appears in 26 - 42% of training articles in politics, economy, health, and history, but in 0 - 1% of test articles in politics, economy, and health. Meanwhile, it appears in 41 - 71% of test articles in sport, entertainment, religion, and relationship, where it is almost absent from training. The official splits therefore appear to be divided by source or collection period, not drawn at random. This explains the distribution shift in Section 3.3, and it means the test set measures generalisation to differently sourced news, a harder and more realistic setting than a random split.

6.5 Likely label noise

The Transformer's most confident errors are three articles labelled *entertainment* and predicted *sport* with 99.4 - 99.5% confidence. As a Kinyarwanda speaker, I read all three as sport articles:

national-team coach Mashami Vincent preparing the Amavubi for a match against Côte d'Ivoire, the USA beating England in a Women's World Cup semi-final, and APR FC winning a league match. In these cases, the model is right, and the label is wrong. A plausible explanation is that some websites publish sport under a combined sport-and-entertainment section, and that section name became the label. Label noise of this kind caps the measurable accuracy of any model and inflates the error count.

6.6 Successful and unsuccessful examples

- **Success:** an IGIHE article from 11 October 2026 about a security investigation in Poland, published six years after the training data and pasted with its social-media sharing text, was classified *as political* with 94.9% confidence.
- **Success:** test articles on sport (Amavubi U20 in Egypt), relationships and religion (a church bishop's statement) were classified correctly with over 99% confidence.
- **Failure (label noise):** the three football articles labelled *entertainment* described in Section 6.5.
- **Failure (out of scope):** an English sentence about a central bank raising interest rates was classified as *economy* with 95% confidence. The topic is plausible, but the model cannot recognise that the input is not Kinyarwanda news, and its confidence gives no warning. A 14-way softmax has no "none of the above" option.

6.7 Discussion

Three conclusions emerge. First, on this task, **data issues dominate model choice**: deduplication, class weighting, and the split shift moved the scores more than the step from logistic regression to a pretrained Transformer. This agrees with MasakhaNEWS, where strong baselines stayed competitive (Adelani et al., 2023). Second, **pretraining on African languages helps, but tokenizer fit is not the whole story**: AfriBERTa tokenises Kinyarwanda more efficiently, yet AfroXLMR's larger capacity generalised better to the shifted test set, consistent with Alabi et al. (2022). Third, **Kinyarwanda's morphology and spelling variation are visible throughout**: character features beat word features, 7.8% of test tokens are unseen, and Word2Vec groups variants such as *abanyeshuri* / *abanyeshuli*. This supports the case for morphology-aware modelling made by Nzeyimana and Niyongabo Rubungo (2022).

7. Deployment

The final model (T3) is publicly usable at kinyarwandanews.streamlit.app: users paste a Kinyarwanda article and receive the predicted category, its confidence, and the top five alternatives.

Architecture. The fine-tuned model and tokenizer are hosted on the Hugging Face Hub as henryparfait/kinnews-topic-classifier, with a model card documenting data, training, results, and limitations. The web app is deployed on Streamlit Community Cloud directly from the GitHub repository. It has two parts:

- `predictor.py` connects the app to the model. It downloads the model from the Hub once (cached for later requests), tokenises the input with truncation at 512 tokens, runs the model on CPU, and returns the softmax probabilities ranked from highest to lowest.
- `streamlit_app.py` is the interface: a text box, a menu of example articles, the prediction with confidence, the top-5 categories as bars, and an "About this model" panel with the test scores and limitations.

Safeguards. The app warns when confidence is below 50%, which Section 6.3 shows corresponds to below 30% accuracy. It notes when an article is longer than the model can read. It also asks for more text when the input is under five words. The example menu deliberately includes confident correct cases, two known errors, and an English sentence, so users can see both strengths and failures.

Hosting decision. I first targeted Hugging Face Spaces, but hosting Gradio apps on its free hardware now requires a paid subscription. Hosting the model on the Hub and the interface on Streamlit Community Cloud kept the system free and public. Free Streamlit apps sleep after a period without visitors and wake in about a minute when opened; the app can also be run locally

with the instructions in the README.

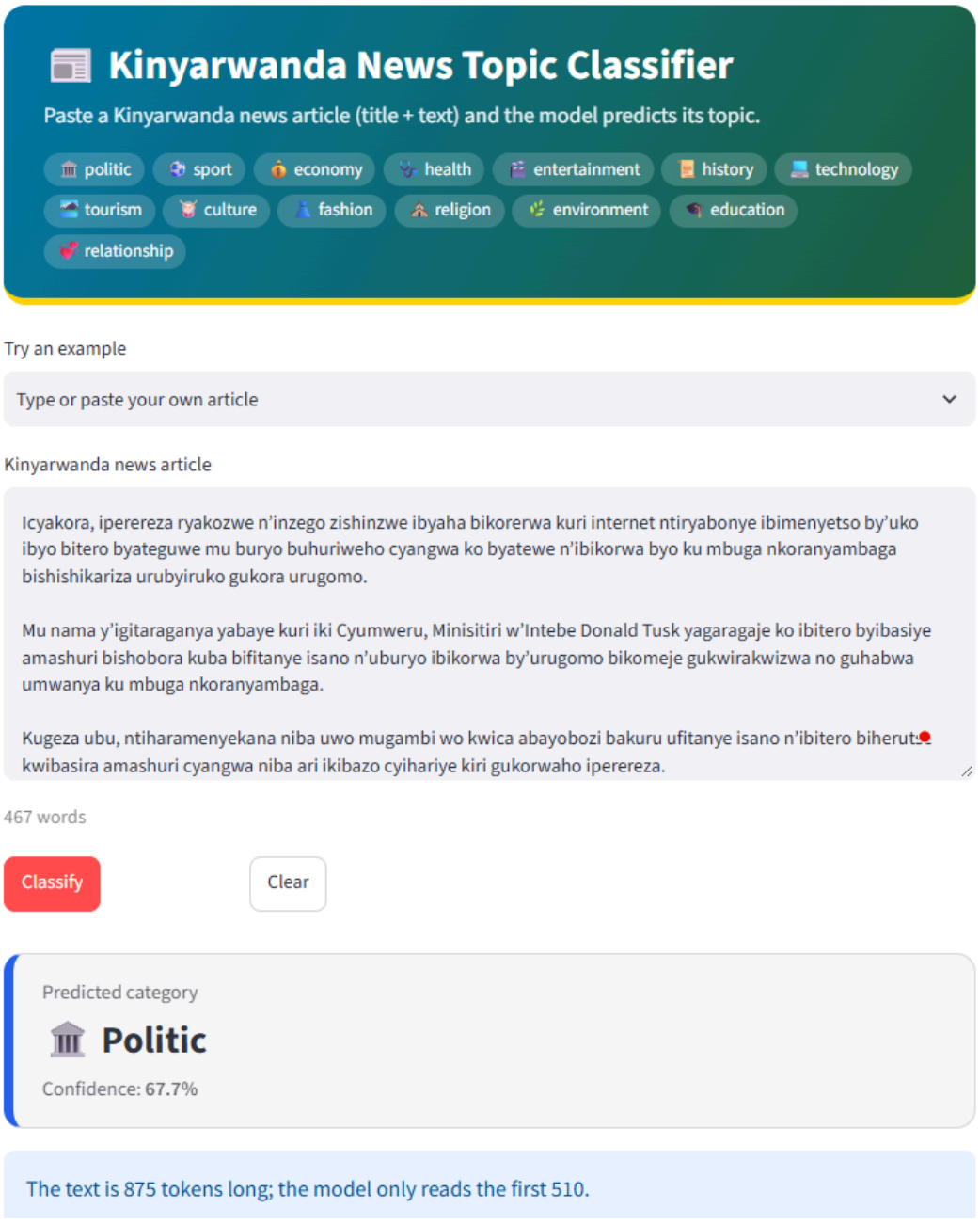


Figure 6: screenshot of the app classifying an article

8. Limitations and Future Work

The main limitations concern the data and the evaluation, not the models, and most could be addressed with more time rather than more computation.

Data. KINNEWS contains label noise (Section 6.5), single labels for multi-topic articles, and news from a limited set of websites around 2019 - 2020, so performance on other sources, later years, or other genres such as social media is unknown. With more time, I would re-annotate a sample of the test set to measure the label-error rate, and allow multiple labels per article.

Evaluation. The validation set has only 3 - 4 articles for some classes, which made model selection noisy (Section 5.3). Each configuration was also trained with a single random seed, so differences of a few hundredths of macro-F1 may not be significant. Future work: a larger, stratified validation set or k-fold cross-validation, and several seeds with confidence intervals.

Models. The hyperparameter search was small (two values of C; one learning rate for the Transformers), and the BiLSTMs were trained without class weights. With more computation, I would fine-tune larger models such as AfroXLMR-large, try KinyaBERT's morphology-aware approach, and apply class weighting to the BiLSTMs for a complete comparison.

Deployment. The app cannot recognise input outside its scope (non-Kinyarwanda or non-news text) and returns a confident prediction anyway (Section 6.7). Adding a language-identification check and an "other" option, and calibrating the confidence scores (for example, with temperature scaling), would make it safer. The free hosting also sleeps when idle and runs on CPU only.

9. Conclusion

I built and deployed a Kinyarwanda news topic classifier whose final model, a fine-tuned AfriBERTa, reaches 75.5% accuracy and 0.581 macro-F1 on a deduplicated KINNEWS test set, and is publicly usable through a web app.

The main findings answer the four questions from the introduction. **(1)** KINNEWS needed substantial cleaning: over half of the training rows were duplicates, 156 articles had conflicting labels, and the official splits differ sharply in class and source distribution. **(2)** Neural sequence models did not automatically beat classical ones: a class-weighted character TF-IDF baseline outperformed every BiLSTM. **(3)** Pretrained African-language Transformers gave the best results, but only modestly; AfroXLMR generalised best (0.618 macro-F1) despite a less efficient Kinyarwanda tokenizer, and reading 512 instead of 256 tokens added almost nothing. **(4)** Remaining errors come mostly from rare and overlapping categories, the source-dependent split, and label noise; the models do not depend on source names, as the shortcut test showed.

The broader lesson for me is that in low-resource NLP, understanding and fixing the data, choosing the right metric, and evaluating honestly mattered more than model size. Character n-grams, class weighting, and a careful look at the errors with a native speaker's eye explained more of the results than any architecture change.

10. References

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Datasets, models and tools

- KINNEWS corpus (MIT license): github.com/Andrews2017/KINNEWS-and-KIRNEWS-Corpus
- AfriBERTa-base: huggingface.co/castorini/afriberta_base
- AfroXLMR-base: huggingface.co/Davlan/afro-xlmr-base
- Streamlit documentation: docs.streamlit.io